

Sharp pointwise bounds for Lebesgue functions

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Abstract

For each n , let λ_n be the Lebesgue function for polynomial interpolation at an arbitrary set of n distinct nodes in $[-1, 1]$. We prove that the points $x \in (-1, 1)$ for which $\lambda_n(x) > (2/\pi) \log n - C(x)$ infinitely often, with some finite $C(x)$, form a dense set. We also prove $\limsup_{n \rightarrow \infty} \lambda_n(x) / \log n \geq 2/\pi$ for almost every $x \in (-1, 1)$. Conversely, we show that no absolute additive constant works for all arrays: for every $M > 0$, there is an array for which each fixed $x \in (-1, 1)$ eventually satisfies $\lambda_n(x) \leq (2/\pi) \log n - M$. For the same array, no single finite constant gives a dense set of points with infinitely many lower-bound occurrences. The positive results use derivative-jump energies and positive Cauchy transforms. The counterexample combines an explicit logarithmic integral estimate on the gaps of a compact set with real-rooted polynomials having prescribed amplitudes.

1 Introduction

For each $n \geq 1$, let

$$-1 \leq x_{1,n} < \cdots < x_{n,n} \leq 1.$$

Define the node polynomial, the cardinal polynomials, and the Lebesgue function by

$$P_n(x) = \prod_{k=1}^n (x - x_{k,n}),$$
$$\ell_{k,n}(x) = \frac{P_n(x)}{(x - x_{k,n})P'_n(x_{k,n})}, \quad \lambda_n(x) = \sum_{k=1}^n |\ell_{k,n}(x)|.$$

The cardinal polynomials are interpreted by continuity at the nodes. The rows of the array need not be nested.

Theorem 1. *For every triangular array of distinct nodes as above, the following statements hold.*

(i) *The set of points $x \in (-1, 1)$ for which there is a finite constant $C(x)$ such that*

$$\lambda_n(x) > \frac{2}{\pi} \log n - C(x) \tag{1.1}$$

for infinitely many n is dense in $(-1, 1)$.

(ii) *For Lebesgue almost every $x \in (-1, 1)$,*

$$\limsup_{n \rightarrow \infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}. \tag{1.2}$$

Theorem 2. *For every $M > 0$, there is a triangular array of distinct nodes in $(-1, 1)$ such that, for every fixed $x \in (-1, 1)$,*

$$\lambda_n(x) \leq \frac{2}{\pi} \log n - M \quad \text{for all sufficiently large } n. \quad (1.3)$$

The starting index may depend on x . The array can also be chosen so that, for every finite C , the set

$$G_C = \left\{ x \in (-1, 1) : \lambda_n(x) > \frac{2}{\pi} \log n - C \text{ for infinitely many } n \right\} \quad (1.4)$$

is not dense in $(-1, 1)$.

Erdős [4, p. 68] asked whether every triangular array has a point at which the sharp logarithmic lower bound holds infinitely often with a bounded additive loss, and whether the normalized lower bound (1.2) holds almost everywhere. Theorem 1(i) answers the first question with a constant that may depend on the point; part (ii) answers the second. As Tao observes in [10, Remark 1.12], the original question does not specify the allowed dependence of the additive constant. Theorem 2 distinguishes the uniform readings from the point-dependent one.

For arbitrary triangular arrays, the three possible quantifier patterns can now be distinguished, using G_C from (1.4) for the array in question:

- (S1) There is an absolute C such that $G_C \neq \emptyset$ for every array. This is false: use Theorem 2 with $M > \max\{C, 0\}$.
- (S2) For every array, there is a finite C , depending on the array but not on the point, such that G_C is dense. This is false by the second assertion of Theorem 2.
- (S3) For every array, there are a finite C and a point $x \in G_C$. This is true by Theorem 1(i). More precisely, $\bigcup_{C \in \mathbb{R}} G_C$ is dense; the constant may vary from point to point in this union.

Theorem 2 concerns arbitrary triangular arrays, as in [4, p. 68] and [10, Corollary 1.11]. The problem page [2] instead uses the initial segments of a single infinite node sequence. Our counterexample does not impose this nesting, and the uniform-constant question in that setting remains open.

1.1 Background

Faber [9] proved the order $\log n$ for the global Lebesgue constant, Bernstein [1] obtained the asymptotically sharp leading constant $2/\pi$, Erdős and Turán [6] improved the error to $O(\log \log n)$, Erdős [5] to $O(1)$, and Vértési [11, Corollary 1] refined the asymptotics through the optimal constant term to an error of $O((\log \log n / \log n)^2)$. See [10, Section 1.3] and [3] for further historical discussion.

For triangular arrays, Erdős [4, p. 68] stated that the set of points x satisfying

$$\limsup_{n \rightarrow \infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}$$

is everywhere dense, and asked whether this holds for almost every x . Erdős and Vértési [7] proved almost-everywhere divergence of interpolants of a suitable continuous function for every array. Their subsequent paper [8, Theorem 2.1] gives a quantitative lower bound for the Lebesgue functions: for every $\varepsilon > 0$, there are $\eta(\varepsilon) > 0$ and $n_0(\varepsilon)$ such that, for every array and $n \geq n_0(\varepsilon)$, there is a measurable set $H_n \subset [-1, 1]$ satisfying

$$|H_n| \leq \varepsilon, \quad \lambda_n(x) > \eta(\varepsilon) \log n \quad (x \in [-1, 1] \setminus H_n).$$

They note that Chebyshev nodes show this order to be optimal. The exceptional sets may vary with n ; the estimate implies $\limsup_n \lambda_n(x)/\log n > 0$ almost everywhere (indeed, $|\bigcup_N \bigcap_{n \geq N} H_n| \leq \varepsilon$, so outside a set of measure at most ε , each point leaves H_n infinitely often; now let $\varepsilon \downarrow 0$). Thus the logarithmic order was already known in 1981. Theorem 1(ii) supplies the optimal coefficient $2/\pi$, which is the coefficient in Erdős’s question and in [10, Remark 1.12]. Erdős and Vértési also give the integral version [8, Corollary 2.2]:

$$\int_{S_n} \lambda_n(x) dx > (|S_n| - \varepsilon) \eta(\varepsilon) \log n \quad (n \geq n_0(\varepsilon))$$

for arbitrary measurable $S_n \subset [-1, 1]$.

Corollary 11 gives, in every sufficiently large row, a point in each fixed positive-measure set with $\lambda_n(x) > c \log n$, for any $0 < c < 2/\pi$. Erdős–Vértési’s theorem instead controls all points outside a small exceptional set, with coefficient $\eta(\varepsilon)$.

Tao [10, Theorem 1.10(i)] proved that every fixed nondegenerate interval $I \subset [-1, 1]$ satisfies

$$\sup_{x \in I} \lambda_n(x) \geq \frac{2}{\pi} \log n - O_I(1).$$

His Corollary 1.11 and Remark 1.12 give, for every prescribed $\omega(n) \rightarrow \infty$, a dense, indeed comeager, set of points x for which

$$\lambda_n(x) \geq \frac{2}{\pi} \log n - \omega(n)$$

infinitely often. Theorem 1(i) replaces $\omega(n)$ by a finite constant depending on the point, on a dense set. In footnote 6 to Remark 1.12 of [10], Tao expresses the tentative belief that “this dependency on I in the error term cannot be entirely eliminated.” Corollary 12 establishes this for the single array of Theorem 2.

1.2 Organization

Sections 2–4 prove Theorem 1(i). The local differentiation estimate turns large derivative jumps into nearby high values of the Lebesgue function. A weighted energy bound supplies enough such jumps, and a second-moment argument gives recurrence on a positive-measure set. Baire’s theorem then gives density.

Sections 5–6 prove Theorem 1(ii) by a separate argument using positive Cauchy transforms. This proof is independent of Sections 2–4 and of Tao’s estimates. Section 7 proves Theorem 2 independently of both lower-bound arguments. Its first three subsections construct positive functions with large logarithmic integral ratios; Lemma 10 converts them into interpolation rows, which are assembled in the final subsection. Section 8 gives the positive-measure and interval-constant corollaries.

Both main theorems and the two corollaries are formalized in Lean 4 with Mathlib. The estimates used from Tao in Proposition 4 are proved directly in Lean. The file `Erdos1132/Theorems.lean` collects the four statements. The `checks/` directory also gives them with explicit node conditions and Lagrange products. The source and build instructions are at <https://github.com/FireflySentinel/erdos-1132>.

Throughout the paper, all logarithms are natural and $|E|$ denotes Lebesgue measure. We write $c_0 = 2/\pi$, and suppress the row index when discussing a single row. Constants in local estimates may depend on the fixed intervals and test functions, but not on n or on interpolation data bounded by one. The notation $J \Subset U$ means that J is a compact subset of the open set U .

2 A local differentiation formula

We use the following local form of Riesz's differentiation formula and its near-equality consequence.

Lemma 3. *Suppose that $T \geq 2\pi$ and F is holomorphic in a neighborhood of the disk $\{|z| \leq T\}$, is real on its real diameter, and satisfies, throughout this disk,*

$$|F(u + iv)| \leq e^{|v|}.$$

Then

$$|F'(0)| \leq 1 + O(T^{-1}). \quad (2.1)$$

If $F'(0) \geq 1 - \delta$, where $\delta \geq 0$, then

$$F(\pi/2) \geq 1 - \frac{\pi^2}{4}\delta - O(T^{-1}). \quad (2.2)$$

The constants are absolute.

Proof. Choose $R = m\pi$ with $T/2 \leq R < T$, and integrate

$$\frac{F(z)}{z^2 \cos z}$$

around $|z| = R$. On this circle, $e^{|\Im z|} \leq 8|\cos z|$. Indeed, for $|v| \geq 1$, use $|\cos(u + iv)| \geq \sinh |v| \geq e^{|v|}/4$. For $|v| \leq 1$, put $d = R - |u| = v^2/(R + |u|) \leq 1/2$; then $|\cos u| = |\cos d| \geq 1/2$, while $e^{|v|} \leq e < 3$. The contour integral, divided by $2\pi i$, has absolute value at most $8/R$. Taking residues at zero and at $t_j = (j + \frac{1}{2})\pi$ gives

$$F'(0) = \sum_{|t_j| < R} \frac{(-1)^j F(t_j)}{t_j^2} + O(T^{-1}). \quad (2.3)$$

Applying the same formula to $F(z) = \sin z$ gives

$$\sum_{|t_j| < R} \frac{1}{t_j^2} = 1 + O(T^{-1}).$$

Together with $|F(t_j)| \leq 1$, equation (2.1) follows. Isolating the term $t_0 = \pi/2$ in (2.3) gives

$$1 - \delta \leq \frac{4}{\pi^2} F(\pi/2) + 1 - \frac{4}{\pi^2} + O(T^{-1}),$$

which proves (2.2). □

We apply this formula to interpolants with bounded nodal data. We first state the local estimates in the form used here, with the normalization:

$$S_n = \sum_k \frac{1}{|P'_n(x_{k,n})|}, \quad \alpha_n = \frac{1}{n} \log S_n, \quad U_n(z) = \frac{1}{n} \log \frac{1}{|P_n(z)|}.$$

Proposition 4 (Tao's local estimates). *Let $I \Subset (-1, 1)$ be a fixed nondegenerate closed interval, and suppose that $\sup_I \lambda_n \leq n$ for all sufficiently large n . This is a specialization of the polynomial upper bound in [10, equation (4.1)]. Define*

$$\rho_n(x) = -\frac{1}{\pi^2} \int_{\mathbb{R} \setminus I} \frac{U_n(y) - \alpha_n}{(y - x)^2} dy.$$

For each fixed compact interval $I' \Subset \text{int } I$, the following hold uniformly for all sufficiently large n :

1. The densities are bounded above and below by positive constants and have uniformly bounded Lipschitz norms:

$$\rho_n(x) \asymp_{I'} 1, \quad |\rho_n(x) - \rho_n(x_0)| \ll_{I'} |x - x_0| \quad (x, x_0 \in I').$$

This is the form of [10, Theorem 4.1(ii), equation (4.6)] needed here.

2. For $x \in I'$ and $\eta \geq 1/n$,

$$U_n(x + i\eta) = \alpha_n - \pi\eta\rho_n(x) + O_{I'}\left(\frac{\log n}{n} + \eta^3\right). \quad (2.4)$$

Compare [10, equation (4.7)].

3. For globally Lipschitz functions h_n supported in I' , with uniformly bounded supremum norms and Lipschitz constants,

$$\frac{1}{n} \sum_k h_n(x_{k,n}) = \int h_n(x) \rho_n(x) dx + O(n^{-1/4}). \quad (2.5)$$

The Lean development proves this proposition directly in the stated form, including the potential expansion (`uniform_complex_potential_expansion`) and density positivity (`eventually_exterior_density_positive`). Thus the formal proof of Theorem 1 does not depend on [10].

The density is defined from I , as in [10, equation (4.15)], and is independent of the auxiliary interval I' . The quadrature estimate follows directly from the Poisson representation: averaging Poisson kernels between heights h and $2h$ gives an approximate identity with Lipschitz error $O(\sqrt{h})$, while (2.4) contributes $O(\log n/(nh) + h^2)$. Taking $h = n^{-1/2}$ gives (2.5).

Lemma 5. *Let $I \subseteq (-1, 1)$ be a fixed nondegenerate closed interval. Suppose that, for all sufficiently large n ,*

$$\lambda_n(x) \leq \Lambda_n := c_0 \log n + C \quad (x \in I), \quad (2.6)$$

where C is fixed. This bound gives the hypothesis of the proposition above; let ρ_n be its densities. For every fixed interval $J \subseteq \text{int } I$ and every real interpolant

$$q(x) = \sum_{k=1}^n a_k \ell_{k,n}(x), \quad |a_k| \leq 1,$$

uniformly for $x \in J$,

$$|q'(x)| \leq \pi n \rho_n(x) (\Lambda_n + o(1)), \quad |q''(x)| \leq C_J n^2 \log n. \quad (2.7)$$

Moreover, for every fixed D , if

$$\frac{|q'(x)|}{\pi n \rho_n(x)} \geq c_0 \log n - D, \quad (2.8)$$

then there is $y = x + O_J(n^{-1})$ such that

$$\lambda_n(y) \geq c_0 \log n - D', \quad (2.9)$$

where D' depends on I, J, C, D , but not on n, x, q . The point y lies in any fixed compact interior interval containing J in its interior, for all sufficiently large n .

Proof. Fix $I' \in \text{int } I$ with $J \in \text{int } I'$. For $\Re z \in I'$ and $\Im z = \eta \geq \log n/n$, the interpolation formula and (2.4) give

$$|q(z)| \leq \frac{S_n |P_n(z)|}{\eta} \leq \frac{1}{\eta} \exp\left(\pi n \eta \rho_n(\Re z) + O(\log n + n \eta^3)\right). \quad (2.10)$$

Fix $x \in J$, and put

$$L = n^{-1/5}, \quad H = n^{-2/5}, \quad \omega = \pi n \rho_n(x)(1 + n^{-1/10}).$$

On the upper edge of $\{|\Re z - x| \leq L, 0 \leq \Im z \leq H\}$, we have $|q(z)| \leq \Lambda_n e^{\omega H}$. ωH exceeds $\pi n H \rho_n(x)$ by $\pi \rho_n(x) n^{1/2}$, which dominates the $O(n^{2/5})$ from the Lipschitz variation of ρ_n and the remaining $O(\log n)$ terms in (2.10). On the lower edge $|q| \leq \Lambda_n$; a Chebyshev expansion on I gives $|q| \leq \Lambda_n \exp(O_I(n))$ on the vertical sides. For $f(z) = \Lambda_n^{-1} q(z) e^{i\omega(z-x)}$, the horizontal edges have $|f| \leq 1$, and the vertical edges have $|f| \leq e^B$ with $B = O_I(n)$. Set

$$\mathcal{B}(z) = 2B \left[e^{(z-x-L)/H} + e^{(-z+x-L)/H} \right].$$

Its real part is nonnegative on the horizontal edges and at least B on the vertical edges, since $\cos(v/H) \geq 1/2$ for $0 \leq v \leq H$. The maximum modulus principle gives $|f(z) e^{-\mathcal{B}(z)}| \leq 1$. At distance at least $L/2$ from the vertical sides, the resulting error factor is $\exp(O(ne^{-L/(2H)})) = 1 + O(n^{-10})$. Thus, also reflecting across the real axis,

$$|q(x + u + iv)| \leq \Lambda_n (1 + n^{-10}) e^{\omega|v|} \quad (|u| \leq L/2, |v| \leq H). \quad (2.11)$$

The rescaled function

$$F(t) = \frac{q(x + t/\omega)}{\Lambda_n (1 + n^{-10})}$$

satisfies Lemma 3 on a disk of radius $T \asymp_J n^{3/5}$. By (2.1),

$$|q'(x)| = \omega \Lambda_n (1 + n^{-10}) |F'(0)| \leq \pi n \rho_n(x) \Lambda_n (1 + n^{-1/10}) (1 + O(n^{-3/5})).$$

Since $\Lambda_n n^{-1/10} = o(1)$, this gives the first bound in (2.7); Cauchy's estimate on a disk of radius comparable to n^{-1} gives the second. Under (2.8), change the sign of q if necessary. For sufficiently large n ,

$$F'(0) \geq \frac{c_0 \log n - D}{\Lambda_n (1 + n^{-1/10}) (1 + n^{-10})} \geq 1 - \frac{C_1}{\Lambda_n}, \quad C_1 = 1 + \max\{C + D, 0\}.$$

Equation (2.2) then gives $|q(x + \pi/(2\omega))| \geq \Lambda_n - C_2$. Since $|q(y)| \leq \lambda_n(y)$ on the real axis and $\omega \asymp_J n$, this proves (2.9) with the asserted displacement. $\square$

3 Derivative jumps and a Riesz energy

At each internal interpolation node, write

$$J_k = \lambda'_n(x_{k,n}+) - \lambda'_n(x_{k,n}-).$$

At the first and last nodes, use the exterior polynomial pieces. Set

$$\nu_k = \frac{1}{S_n |P'_n(x_{k,n})|}.$$

The absolute values of all cardinal polynomials other than $\ell_{k,n}$ have a cusp at $x_{k,n}$, whereas $\ell_{k,n}$ is positive near that node. Using $\ell_{j,n}(x) = P_n(x)/((x-x_{j,n})P'_n(x_{j,n}))$ and $P_n(x_{k,n}) = 0$ to differentiate at $x_{k,n}$, we obtain

$$J_k = 2 \sum_{j \neq k} |\ell'_{j,n}(x_{k,n})| = 2 \sum_{j \neq k} \frac{\nu_j}{\nu_k |x_{k,n} - x_{j,n}|}. \quad (3.1)$$

Lemma 6. *Under the hypotheses of Lemma 5, choose a nonzero nonnegative smooth function f compactly supported in $\text{int } I$. Then*

$$\sum_k \frac{f(x_{k,n})^2}{n \rho_n(x_{k,n})} \frac{J_k}{2\pi n \rho_n(x_{k,n})} \geq c_0 \log n \int f(x)^2 dx - O(1). \quad (3.2)$$

Terms with $f(x_{k,n}) = 0$ are taken to be zero; ρ_n is only needed on a fixed interior neighborhood of the support of f .

Proof. Put $a_k = f(x_{k,n})/\rho_n(x_{k,n})$, with $a_k = 0$ outside the support of f , and define

$$\mu_n = \frac{1}{n} \sum_k a_k \delta_{x_{k,n}}, \quad K_\varepsilon(t) = \frac{1}{\sqrt{t^2 + \varepsilon^2}}.$$

The kernel K_ε is positive definite, since

$$K_\varepsilon(t) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-s\varepsilon^2} e^{-st^2} ds$$

is a positive mixture of Gaussian kernels. Write

$$\mathcal{E}_\varepsilon(\sigma, \tau) = \iint K_\varepsilon(x - y) d\sigma(x) d\tau(y)$$

for its bilinear energy. Positivity applied to $\mu_n - f(x) dx$ gives

$$\mathcal{E}_\varepsilon(\mu_n, \mu_n) \geq 2\mathcal{E}_\varepsilon(\mu_n, f dx) - \mathcal{E}_\varepsilon(f dx, f dx). \quad (3.3)$$

Take $\varepsilon = 1/n$. Subtracting $f(x)$ in the integral over $|t| \leq 1$, and using the Lipschitz bound for f , gives the uniform estimate

$$(K_{1/n} * f)(x) = 2 \log n f(x) + O_f(1). \quad (3.4)$$

The mass of μ_n is bounded. Also, by (2.5),

$$\int f d\mu_n = \frac{1}{n} \sum_k \frac{f(x_{k,n})^2}{\rho_n(x_{k,n})} = \int f^2 + O(n^{-1/10}). \quad (3.5)$$

Again by (3.4),

$$\mathcal{E}_{1/n}(f dx, f dx) = 2 \log n \int f^2 + O(1).$$

Together with (3.3)–(3.5), this yields

$$\mathcal{E}_{1/n}(\mu_n, \mu_n) \geq 2 \log n \int f^2 - O(1). \quad (3.6)$$

The diagonal part of this energy is $n^{-1} \sum_k a_k^2 = O(1)$. Since $K_{1/n}(t) \leq 1/|t|$ for $t \neq 0$,

$$\frac{1}{n^2} \sum_{k \neq j} \frac{a_k a_j}{|x_{k,n} - x_{j,n}|} \geq 2 \log n \int f^2 - O(1). \quad (3.7)$$

Finally,

$$a_k^2 \frac{\nu_j}{\nu_k} + a_j^2 \frac{\nu_k}{\nu_j} \geq 2a_k a_j.$$

By (3.1), the left side of (3.2) is

$$\frac{1}{\pi n^2} \sum_{k \neq j} \frac{a_k^2 \nu_j}{\nu_k |x_{k,n} - x_{j,n}|}.$$

Pairing the terms indexed by (k, j) and (j, k) , and then applying (3.7), bounds this expression below by

$$\frac{1}{\pi n^2} \sum_{k \neq j} \frac{a_k a_j}{|x_{k,n} - x_{j,n}|} \geq \frac{2}{\pi} \log n \int f^2 - O(1).$$

This proves (3.2). □

4 A dense set of points with bounded additive loss

Proposition 7. *Assume the local upper bound (2.6) on a fixed nondegenerate interval $I \Subset (-1, 1)$. There are a constant D and a measurable set $G \subset \text{int } I$, with $|G| > 0$, such that every $x \in G$ satisfies*

$$\lambda_n(x) \geq c_0 \log n - D$$

for infinitely many n .

Proof. Choose compact intervals $J_0 \Subset J_1 \Subset \text{int } I$, and choose f as in Lemma 6 with $\text{supp } f \subset J_0$. All constants below are independent of n : after I, C, J_0, J_1, f are fixed, we choose D , then $\kappa, D_1, d, \kappa_1, d_1$, and finally b, D_2 . Let

$$w_k = \frac{f(x_{k,n})^2}{n \rho_n(x_{k,n})}, \quad T_k = \frac{J_k}{2\pi n \rho_n(x_{k,n})}, \quad W = \int f^2 > 0.$$

By (2.5),

$$\sum_k w_k = W + O(n^{-1/10}), \quad 0 \leq w_k \leq C_f/n. \quad (4.1)$$

On each interval between successive nodes, λ_n equals an interpolant with nodal data in $\{-1, 1\}$. Both one-sided derivatives at a node therefore satisfy (2.7), so

$$0 \leq T_k \leq \Lambda_n + o(1) \quad (4.2)$$

uniformly on the support of f . Lemma 6, (4.1), and (4.2) show that

$$\sum_k w_k (\Lambda_n + 1 - T_k) = O(1), \quad (4.3)$$

with nonnegative summands for all sufficiently large n . On a node with $T_k < c_0 \log n - D$, the factor in (4.3) is greater than $C + 1 + D$. Hence, by choosing D sufficiently large, such nodes carry

at most $W/2$ of the weights. Since (4.1) has total weight $W + o(1)$, for all sufficiently large n the nodes in the support of f satisfying

$$T_k \geq c_0 \log n - D \quad (4.4)$$

carry at least $W/4$ of the weights. In particular, there are at least κn such nodes, with $\kappa > 0$ independent of n . Discarding the first and last nodes does not change this conclusion after reducing κ .

At a node satisfying (4.4), at least one adjacent polynomial piece q has

$$|q'(x_{k,n})| \geq \pi n \rho_n(x_{k,n})(c_0 \log n - D).$$

Lemma 5 gives a point

$$y_k = x_{k,n} + O(n^{-1}), \quad \lambda_n(y_k) \geq c_0 \log n - D_1. \quad (4.5)$$

The same polynomial piece q has value 1 at both endpoints of the corresponding gap. Rolle's theorem and (2.7) imply that this gap has length at least d/n for some fixed $d > 0$ when the gap is contained in J_1 . If it is not contained in J_1 , its length is bounded below by $\text{dist}(J_0, \mathbb{R} \setminus J_1) > 0$.

An interval of length less than d/n contains at most two such nodes, so every interval of length $O(n^{-1})$ contains only a bounded number of the nodes in (4.4). By (4.5), the same holds, with multiplicities, for the points y_k . Hence one can select at least $\kappa_1 n$ of them, separated by d_1/n , for fixed $\kappa_1, d_1 > 0$.

Equation (2.7), applied to every polynomial piece, also shows that λ_n is $C_I n \log n$ -Lipschitz on a fixed compact interior interval containing these points. Choose a fixed sufficiently small $b > 0$, and let A_n be the union of the intervals of radius

$$r_n = \frac{b}{n \log n}$$

centered at the selected points. For all sufficiently large n , these intervals are disjoint, lie in a fixed compact interior interval, and satisfy

$$\lambda_n(x) \geq c_0 \log n - D_2 \quad (x \in A_n), \quad \frac{c}{\log n} \leq |A_n| \leq \frac{C_2}{\log n}. \quad (4.6)$$

The separation of the centers in the m -th row implies, for every interval B ,

$$|A_m \cap B| \leq C_3 \left(\frac{|B|}{\log m} + \frac{1}{m \log m} \right).$$

Summing this inequality over the at most n components of A_n gives, for $m > n$,

$$|A_n \cap A_m| \leq C_3 \left(\frac{|A_n|}{\log m} + \frac{n}{m \log m} \right). \quad (4.7)$$

Put $B_j = A_{2^j}$. Equations (4.6) and (4.7) give

$$|B_j| \asymp j^{-1}, \quad |B_j \cap B_k| \leq C_4 \left(\frac{1}{jk} + \frac{2^{j-k}}{k} \right) \quad (j < k). \quad (4.8)$$

For every fixed j_0 , the sum of the measures of B_j , $j_0 \leq j \leq N$, grows at least as a positive constant times $\log N$. By (4.8), the integral of the square of the sum of their indicators is $O((\log N)^2)$. Cauchy-Schwarz consequently gives a positive lower bound, independent of j_0 , for

$$\left| \bigcup_{j \geq j_0} B_j \right|.$$

Continuity of measure along these decreasing tail unions shows that

$$|\limsup_{j \rightarrow \infty} B_j| > 0. \quad (4.9)$$

Together with (4.6), this proves the proposition. $\square$

Proof of Theorem 1(i). Fix a nondegenerate compact interval $K_0 \subset (-1, 1)$, and suppose that it contains no point satisfying the assertion. Then

$$\lambda_n(x) - c_0 \log n \longrightarrow -\infty \quad \text{for every } x \in K_0. \quad (4.10)$$

The closed sets

$$F_M = \{x \in K_0 : \lambda_n(x) \leq c_0 \log n + M \text{ for every } n \geq 2\}, \quad M = 1, 2, \dots,$$

cover K_0 , by (4.10). Baire's theorem supplies a nondegenerate closed subinterval $I \subset K_0$ on which (2.6) holds with a fixed C . Proposition 7 contradicts (4.10). Thus every such K_0 contains a good point, proving density. Increasing the final additive constant by one gives the strict inequality in the statement. $\square$

5 Positive Cauchy transforms

Suppress the row index and put

$$S = \sum_k \frac{1}{|P'(x_k)|}, \quad \nu_k = \frac{1}{S|P'(x_k)|}, \quad \epsilon_k = \operatorname{sgn} P'(x_k).$$

Define

$$m(z) = \sum_k \frac{\nu_k}{z - x_k}, \quad g(z) = \sum_k \frac{\epsilon_k \nu_k}{z - x_k} = \frac{1}{SP(z)}. \quad (5.1)$$

The last identity is the partial fraction formula for $1/P$, whose expansion at infinity gives

$$\sum_k \epsilon_k \nu_k x_k^j = 0 \quad (0 \leq j \leq n-2). \quad (5.2)$$

For $n \geq 2$, both $m + g$ and $m - g$ are Cauchy transforms of positive probability measures: their masses at x_k are $(1 \pm \epsilon_k) \nu_k$.

For real x away from the nodes, write

$$H(x) = \sum_k \frac{\nu_k}{|x - x_k|}, \quad \gamma_s(x) = -\Im m(x + is), \quad p_s(t) = \frac{s}{\pi(t^2 + s^2)}.$$

Then

$$\lambda_n(x) = \frac{H(x)}{|g(x)|}, \quad H(x) = \frac{2}{\pi} \int_0^\infty \gamma_s(x) \frac{ds}{s}, \quad (5.3)$$

and the Poisson semigroup gives

$$\gamma_s = p_{s-h} * \gamma_h \quad (s > h). \quad (5.4)$$

Lemma 8. *For every row with $n \geq 2$, every $0 < h \leq 1$, and every $x \in [-1, 1]$,*

$$\frac{|g(x + ih)|}{\gamma_h(x)} \leq \frac{4 + h^2}{h^2 \sinh((n-1)h/2)}. \quad (5.5)$$

Consequently, if $0 < a < 1$ and $h = n^{-a}$, then

$$\sup_{x \in [-1, 1]} \frac{|g(x + ih)|}{\gamma_h(x)} \longrightarrow 0. \quad (5.6)$$

Proof. Set $z = x + ih$ and $p = T_{n-1}$, the Chebyshev polynomial of the first kind. The polynomial $(p(t) - p(z))/(t - z)$ has degree at most $n - 2$, so (5.2) gives

$$p(z)g(z) = \sum_k \frac{\epsilon_k \nu_k p(x_k)}{z - x_k}, \quad |p(z)g(z)| \leq \frac{1}{h},$$

using $|p(x_k)| \leq 1$ and $\sum_k \nu_k = 1$. Write $z = \cos(u + iv)$, with $u, v \in \mathbb{R}$. Then $h \leq \sinh |v|$, so $|v| \geq \operatorname{arsinh} h \geq h/2$. The identity $T_{n-1}(\cos w) = \cos((n-1)w)$ yields

$$|p(z)|^2 = \cos^2((n-1)u) + \sinh^2((n-1)v) \geq \sinh^2((n-1)h/2).$$

Finally, since the nodes and x lie in $[-1, 1]$,

$$\frac{h}{4 + h^2} \leq \gamma_h(x) = \sum_k \frac{\nu_k h}{(x - x_k)^2 + h^2}.$$

Combining these bounds proves (5.5). For $h = n^{-a}$, its right-hand side tends to zero: the hyperbolic sine grows exponentially in n^{1-a} , while h^{-2} grows polynomially. This proves (5.6). $\square$

Lemma 9. *Let F be the Cauchy transform of a positive probability measure with finite support. For every bounded real interval S_0 , every $x \in \mathbb{R}$, and every $h > 0$,*

$$\int_{\mathbb{R}} p_h(x - y) \mathbf{1}_{\{F(y) \in S_0\}} dy = \frac{1}{\pi} \int_{S_0} \frac{-\Im F(x + ih)}{(t - \Re F(x + ih))^2 + (\Im F(x + ih))^2} dt. \quad (5.7)$$

Proof. The function F maps the upper half-plane into the lower half-plane. Compose harmonic measure of S_0 there with F . The resulting bounded harmonic function on the upper half-plane has the indicated boundary values almost everywhere. The exceptional boundary points are poles of F and preimages of the endpoints of S_0 , a finite set. Its Poisson representation is (5.7). $\square$

6 The almost-everywhere bound

Proof of Theorem 1(ii). The failure set is the union, over rational $0 < c < c_0$ and integers $N \geq 2$, of the measurable sets

$$\{y \in (-1, 1) : \lambda_n(y) \leq c \log n \text{ for every } n \geq N\}.$$

If the assertion fails, one of these sets E has positive measure. For its fixed c, N , we therefore have

$$\lambda_n(y) \leq c \log n \quad (y \in E, n \geq N). \quad (6.1)$$

Choose constants

$$K > 1, \quad 0 < b < a < 1, \quad \frac{2b}{\pi cK} > 1,$$

which is possible because $c < 2/\pi$. Set

$$h = n^{-a}, \quad \eta = n^{-b}, \quad H^\eta(x) = \frac{2}{\pi} \int_\eta^1 \gamma_s(x) \frac{ds}{s}.$$

By (5.3), $H^\eta \leq H$. For $s > 0$ and real x, y, t , the triangle inequality gives

$$|y + is - t| \leq |x + is - t| + |x - y| \leq \left(1 + \frac{|x - y|}{s}\right) |x + is - t|.$$

Taking reciprocal squares and summing against the positive masses yields $\gamma_s(x) \leq (1 + |x - y|/s)^2 \gamma_s(y)$. Integrating over $\eta \leq s \leq 1$, we obtain

$$H^\eta(x) \leq \left(1 + \frac{|x - y|}{\eta}\right)^2 H^\eta(y). \quad (6.2)$$

Fix $x \in E$, and write

$$u = \Re m(x + ih), \quad \gamma = \gamma_h(x), \quad S_x = [u - K\gamma, u + K\gamma].$$

By (5.6) and (5.7), each of the two events

$$m(y) + g(y) \in S_x, \quad m(y) - g(y) \in S_x$$

has probability

$$p_K + o(1), \quad p_K = \frac{2}{\pi} \arctan K > \frac{1}{2},$$

under the Poisson probability $p_h(x - y) dy$. The error is uniform in $x \in E$: for $F = m \pm g$, (5.6) gives $\Re F(x + ih) = u + o(\gamma)$ and $-\Im F(x + ih) = \gamma(1 + o(1))$. Substituting $t = u + \gamma s$ in (5.7) therefore gives $\pi^{-1} \int_{-K}^K (1 + s^2)^{-1} ds + o(1) = p_K + o(1)$. The two events have intersection probability at least

$$q_K + o(1), \quad q_K = 2p_K - 1 > 0. \quad (6.3)$$

Choose R so that the Poisson mass of $|y - x| > Rh$ is less than $q_K/8$, and let

$$E_n = \{x \in E : (p_h * \mathbf{1}_{\mathbb{R} \setminus E})(x) < q_K/8\}.$$

The approximate identity theorem gives $|E \setminus E_n| \rightarrow 0$. For all sufficiently large n , (6.3) is greater than $q_K/2$. For $x \in E_n$, the two excluded sets $\mathbb{R} \setminus E$ and $\{|y - x| > Rh\}$ have total Poisson mass less than $q_K/4$. Hence there exists $y \in E$ with $|y - x| \leq Rh$ and both events true. We may take y away from the nodes. Since $m(y) \pm g(y)$ both lie in an interval of length $2K\gamma_h(x)$,

$$|g(y)| \leq K\gamma_h(x).$$

Equations (5.3) and (6.1) give

$$H^\eta(y) \leq H(y) = \lambda_n(y) |g(y)| \leq cK(\log n) \gamma_h(x).$$

Together with (6.2), this yields

$$H^\eta(x) \leq (1 + Rh/\eta)^2 H^\eta(y) \leq cK(1 + Rh/\eta)^2 (\log n) \gamma_h(x) \quad (x \in E_n). \quad (6.4)$$

Let T_n be convolution with the symmetric probability kernel

$$k_n(t) = \frac{1}{b \log n} \int_{\eta}^1 p_{s-h}(t) \frac{ds}{s}.$$

Its integral is one, since $\int_{\eta}^1 ds/s = b \log n$. Equation (5.4) then shows that

$$H^{\eta} = \frac{2b}{\pi} (\log n) T_n \gamma_h.$$

Thus (6.4) becomes

$$\frac{T_n \gamma_h(x)}{\gamma_h(x)} \leq \frac{1}{A_n} \quad (x \in E_n), \quad A_n = \frac{2b}{\pi c K} (1 + Rh/\eta)^{-2} \longrightarrow A > 1. \quad (6.5)$$

Integrate (6.5) over E_n , discard the contribution from $y \notin E_n$, and symmetrize. Positivity of γ_h , symmetry of k_n , and $(r + r^{-1})/2 \geq 1$ give

$$\frac{|E_n|}{A_n} \geq \iint_{E_n \times E_n} k_n(x-y) \frac{\gamma_h(y)}{\gamma_h(x)} dy dx \geq \iint_{E_n \times E_n} k_n(x-y) dy dx. \quad (6.6)$$

The kernels k_n form an approximate identity. In fact, for every fixed $\delta > 0$,

$$\int_{|t|>\delta} k_n(t) dt \leq \frac{C}{b \log n} \int_{\eta}^1 \min(1, s/\delta) \frac{ds}{s} \longrightarrow 0.$$

It follows that

$$\iint_{E \times E} k_n(x-y) dy dx \longrightarrow |E|.$$

Replacing E by E_n changes this integral by at most $2|E \setminus E_n|$. Taking limits in (6.6) therefore yields

$$\frac{|E|}{A} \geq |E|,$$

contrary to $|E| > 0$ and $A > 1$. This proves the assertion. $\square$

7 Failure of uniform additive constants

We prove Theorem 2. The constants are chosen for a clean verification and are not optimized. Smooth on $I = [-1, 1]$ will mean extendible to a smooth function on a neighborhood of I .

7.1 The compact set

For $I = [-1, 1]$, write

$$Lf(x) = \int_I \frac{f(x) - f(y)}{|x - y|} dy.$$

In particular,

$$\|Lf\|_{\infty} \leq 2\|f'\|_{\infty} \quad (f \in C^1(I)). \quad (7.1)$$

Fix $A > 1$, and put

$$\rho = e^{-1024(A+40)}, \quad \delta_k = \rho 2^{-k}, \quad m_k = \lceil e^{1/\delta_k} \rceil.$$

Starting with I , replace every retained interval of length ℓ_{k-1} at stage k by m_k equally long closed intervals, retaining both endpoints of the parent. The children and intervening open gaps have respective lengths

$$\ell_k = \frac{(1 - \delta_k)\ell_{k-1}}{m_k}, \quad b_k = \frac{\delta_k\ell_{k-1}}{m_k - 1}, \quad \ell_0 = 2.$$

Let F be the intersection of the retained sets and $U = I \setminus F$. Thus F is compact, contains both endpoints, and has empty interior, since $\ell_k \rightarrow 0$. The set U is a dense open union of disjoint gaps, and

$$|U| \leq 2 \sum_k \delta_k = 2\rho. \quad (7.2)$$

We need the following two properties:

$$|F \cap (x - r, x + r)| \geq \kappa r \quad (x \in F, \ 0 < r \leq 1), \quad \kappa = 1/32, \quad (7.3)$$

and

$$\int_U \frac{dy}{|x - y|} = \infty \quad (x \in F). \quad (7.4)$$

For (7.3), choose the first k for which $\ell_k \leq r/16$, and let J be the stage $k - 1$ parent containing x , with $|J| > r/16$. At least half the length of any retained interval belongs to F , since every subsequent product of retained proportions is at least $1 - \rho > 1/2$. If $|J| \leq r$, then $J \subset [x - r, x + r]$, proving (7.3). Otherwise $J \cap [x - r, x + r]$ has length $s \geq r$. Write $a = \ell_k$, $b = b_k$, $\delta = \delta_k$. Then

$$b \leq a, \quad b/(a + b) \leq 2\delta.$$

The gaps intersecting an interval of length s have total length at most $2\delta s + 2b$. The two partially intersected children have total length at most $2a$. The full children inside this interval therefore have total length at least

$$(1 - 2\delta)s - 2(a + b) \geq r/2.$$

Their intersections with F have total length at least $r/4$, as required.

For (7.4), in the stage $k - 1$ parent containing x , one side of its child contains at least $(m_k - 1)/2$ gaps. Use the first $q = \lfloor m_k/8 \rfloor$ on that side. The far endpoint of the j -th such gap is at distance at most $4j\ell_{k-1}/m_k$ from x , and its length is at least $\delta_k\ell_{k-1}/m_k$. These gaps contribute at least

$$\frac{\delta_k}{4} \sum_{j=1}^q \frac{1}{j} \geq \frac{\delta_k}{4} (\log m_k - \log 16) \geq \frac{1}{8}.$$

The gaps used at different stages are disjoint. Summation proves (7.4).

7.2 An explicit logarithmic integral estimate

On each gap (b, c) , define

$$r(x) = \frac{(x - b)(c - x)}{c - b},$$

and put $r = 0$ on F . If $D(x) = \text{dist}(x, F)$, then

$$r(x) \leq D(x) \leq 2r(x) \quad (x \in U). \quad (7.5)$$

The function r is globally 1-Lipschitz: its derivative has absolute value at most one on each gap, and points in distinct gaps can be compared through a point of F between them.

Put

$$B = e^8, \quad \alpha = 1/256, \quad p(t) = [\log(B/t)]^{-\alpha}, \quad u(x) = p(r(x)) \ (x \in U), \quad u = 0 \ (x \in F). \quad (7.6)$$

Then u is continuous on I , bounded, positive and smooth in each gap, and vanishes on F . In particular, $Lu(x)$ is an absolutely convergent integral for every $x \in U$.

We prove the quantitative bound

$$\frac{Lu(x)}{u(x)} \geq \frac{1}{512} \log \frac{1}{4r(x)} - 32 \quad \text{if } \log(B/r(x)) \geq 16. \quad (7.7)$$

All estimates below apply uniformly up to the endpoints of I .

Fix such an x , and write $s = r(x)$, $T = \log(B/s)$, $a_0 = 4s$. For $t \in [0, 2]$, let

$$M_I(t) = |I \cap (x - t, x + t)|, \quad M_U(t) = |U \cap (x - t, x + t)|.$$

Choose $z \in F$ with $|x - z| = D(x) \leq 2s$. If $4s \leq t \leq 2$, the ball of radius $t/2$ about z , intersected with I , is contained in the ball of radius t about x . By (7.3), the latter contains at least $\kappa t/2$ of F . Since $M_I(t) \leq 2t$,

$$M_U(t) \leq (1 - \eta)M_I(t), \quad \eta = \kappa/4 = 1/128 \quad (4s \leq t \leq 2). \quad (7.8)$$

First consider $|y - x| < 4s$. For $|y - x| < s/2$, Lipschitz continuity of r gives $r(y) \in [s/2, 3s/2]$. On this range $p'(r(y)) \leq 4p(s)/s$, by the derivative formula for p and $T \geq 16$. The integral of the absolute difference quotient over this part is thus at most $4p(s)$. On $s/2 \leq |y - x| < 4s$, we have $r(y) \leq 5s$ and

$$p(5s)/p(s) = \left(\frac{T}{T - \log 5} \right)^\alpha \leq 2.$$

The absolute difference is at most $3p(s)$, so this part contributes at most $6 \log 8 p(s) < 18p(s)$. In particular, the whole near contribution to $Lu(x)$ is at least $-24p(s)$.

For the far contribution, set $l_- = 1 + x$, $l_+ = 1 - x$, and

$$H = \sum_{\nu \in \{-, +\}} \max\{\log(l_\nu/(4s)), 0\}.$$

The positive part is exactly $p(s)H$. For the negative part, $r(y) \leq s + |x - y| \leq (5/4)|x - y|$, so it is bounded by

$$\int_{a_0}^2 f(t) dM_U(t), \quad f(t) = \frac{p(5t/4)}{t}.$$

The function f is decreasing. Stieltjes integration by parts, dropping the nonpositive lower-end boundary term and using (7.8), gives

$$\int_{a_0}^2 f(t) dM_U(t) \leq (1 - \eta) \left[f(a_0)M_I(a_0) + \int_{a_0}^2 f(t) dM_I(t) \right]. \quad (7.9)$$

The first term on the right, including its factor, is at most $4p(s)$.

Since $dM_I(t) = 1_{\{t < l_-\}} dt + 1_{\{t < l_+\}} dt$, the remaining term splits into at most two integrals. For a side with $l_\nu > 4s$, put

$$h = \log(l_\nu/(4s)), \quad T_0 = T - \log 5.$$

Then $0 \leq h < T_0$, since $T_0 - h = \log(4B/(5l_\nu)) > 0$, and

$$\int_{4s}^{l_\nu} \frac{p(5t/4)}{t} dt = \int_0^h (T_0 - z)^{-\alpha} dz \leq \frac{T_0^{-\alpha}}{1 - \alpha} h. \quad (7.10)$$

The last inequality follows from $(1 - z)^{1-\alpha} \geq 1 - z$ for $0 \leq z \leq 1$.

Finally, concavity of the power α , $\log 5 < 2$, and $T \geq 16$ give

$$\begin{aligned} \frac{1 - \eta T_0^{-\alpha}}{1 - \alpha p(s)} &= \frac{254}{255} \left(\frac{T}{T - \log 5} \right)^{1/256} \\ &\leq \frac{254}{255} (8/7)^{1/256} \leq \frac{254}{255} \left(1 + \frac{1}{1792} \right) \leq 1 - \frac{1}{512}. \end{aligned} \quad (7.11)$$

Thus the negative far contribution is at most $4p(s) + (1 - 1/512)p(s)H$. Including the near part gives $Lu(x)/u(x) \geq H/512 - 28$. At least one of l_-, l_+ is at least one, so $H \geq \log(1/(4s))$. This proves (7.7), with room in its constant.

Every gap has length at most $|U| \leq 2\rho$, so $r(x) \leq \rho/2$. Our choice of ρ ensures $T \geq 16$, and (7.7) yields

$$\frac{Lu(x)}{u(x)} \geq 2(A + 40) - \frac{\log 2}{512} - 32 > A + 2 \quad (x \in U). \quad (7.12)$$

It also shows that $Lu(x)/u(x) \rightarrow +\infty$ as x approaches either endpoint of any fixed gap.

7.3 Smooth positive functions on the whole interval

Let K_N be the union of the closed middle halves of all gaps removed in the first N stages. This is a compact subset of U . For any smooth $0 \leq \chi \leq 1$, vanishing near F and equal to one on K_N , put

$$A_\chi(x) = \int_U \frac{\chi(y)}{|x - y|} dy \quad (x \in F).$$

The estimate proving (7.4), restricted to these middle halves, gives at least $1/16$ from each of the first N stages. Consequently,

$$A_\chi(x) \geq N/16 \quad (x \in F). \quad (7.13)$$

Let $\epsilon_j = 1/j$, $j \geq 2$, and put

$$d_j = \exp(8 - (2j)^{256}), \quad C_j = 2(2j)^{256}.$$

Since $r \leq D$ and $p(d_j) = 1/(2j)$, one has $u(y) \leq \epsilon_j/2$ when $D(y) \leq d_j$. Using this for $|x - y| < d_j$, and $u \leq 1$ on the remainder, gives

$$\int_U \frac{\chi(y)u(y)}{|x - y|} dy \leq \frac{\epsilon_j}{2} A_\chi(x) + C_j \quad (x \in F). \quad (7.14)$$

Here the far integral is at most $2 \log(2/d_j) \leq C_j$.

Set

$$k_j = \lceil 32(A + 1) \rceil + 64j(2j)^{256} + j.$$

Choose a smooth function $f : \mathbb{R} \rightarrow [0, 1]$ with $\{f > 0\} = U$, and a nondecreasing smooth $q_0 : \mathbb{R} \rightarrow [0, 1]$, equal to zero on $(-\infty, 1]$ and one on $[2, \infty)$. By compactness, choose $0 < \tau_j \leq 1/j$ such that $2\tau_j \leq f$ on K_{k_j} , and define

$$\chi_j(x) = q_0(f(x)/\tau_j).$$

Thus χ_j is smooth, vanishes on a neighborhood of F , and equals one on K_{k_j} . Since $\tau_j \rightarrow 0$, at each fixed point of U it is eventually identically one on a neighborhood of that point. Equation (7.13) gives

$$\frac{\epsilon_j}{2} A_{\chi_j}(x) - C_j \geq \frac{k_j}{32j} - C_j \geq (A+1)\epsilon_j \quad (x \in F).$$

Define

$$v_j = \chi_j u + (1 - \chi_j)\epsilon_j. \quad (7.15)$$

This function is strictly positive and smooth on a neighborhood of I : the function u is smooth and positive in each gap, and $v_j = \epsilon_j$ near F .

At $x \in F$, (7.14) gives $Lv_j(x) \geq (A+1)v_j(x)$. At every fixed $x \in U$, eventually $v_j = u$ on a neighborhood of x . Also $v_j \rightarrow u$ in $L^1(I)$, by bounded convergence. The integrals outside that neighborhood then converge, so

$$\frac{Lv_j(x)}{v_j(x)} \rightarrow \frac{Lu(x)}{u(x)} > A+2 \quad (x \in U). \quad (7.16)$$

We have therefore constructed positive smooth functions such that

$$\frac{Lv_j(x)}{v_j(x)} \geq A+1 \quad \text{eventually for every fixed } x \in I. \quad (7.17)$$

The eventual index is allowed to depend on x .

7.4 From amplitudes to interpolation nodes

We prove an upper estimate with an absolute additive constant.

Lemma 10. *Let $h(z) = \sum_{r=0}^d h_r z^r$ have real coefficients, no zero on the closed unit disk, and $h(0) > 0$. Set*

$$v(\cos \theta) = |h(e^{i\theta})|^{-1}.$$

In this subsection, T_k denotes the Chebyshev polynomial of the first kind, defined by $T_k(\cos \theta) = \cos(k\theta)$. For $n > d$, define the real polynomial

$$P_n(x) = \sum_{r=0}^d h_r T_{n-r}(x). \quad (7.18)$$

For all sufficiently large n , this polynomial has exactly n simple roots in $(-1, 1)$. The corresponding Lebesgue function satisfies, on any fixed $K \Subset (-1, 1)$,

$$\lambda_n(x) \leq \frac{2}{\pi} \log n + 3 - \frac{Lv(x)}{\pi v(x)} \quad (x \in K) \quad (7.19)$$

for all sufficiently large n , with the threshold allowed to depend on h, K .

Proof. The analytic logarithm of h , chosen real at zero, is real on the real diameter. Its imaginary part $\psi(\theta) = \arg h(e^{i\theta})$ thus satisfies $\psi(0) = \psi(\pi) = 0$. We have the exact identity

$$P_n(\cos \theta) = |h(e^{i\theta})| \cos(n\theta - \psi(\theta)).$$

For $n > \|\psi'\|_\infty$, the phase increases strictly from 0 to $n\pi$. The polynomial has degree exactly n , since its leading term comes from $h_0 T_n$, with $h_0 > 0$. This gives the asserted n simple roots. At their angles θ_k ,

$$|P'_n(\cos \theta_k)| = |h(e^{i\theta_k})| \frac{n - \psi'(\theta_k)}{\sin \theta_k}. \quad (7.20)$$

Put $s = (n\theta - \psi(\theta))/n$, let $\theta = \Theta_n(s)$ be the inverse, and set $X_n(s) = \cos \Theta_n(s)$. The roots correspond to $s_k = (k - \frac{1}{2})\pi/n$. Moreover $X_n \rightarrow \cos$ in $C^3([0, \pi])$. At a point $x = X_n(s)$ other than a node, put $b = |\cos(ns)|$. Formula (7.20) gives

$$\lambda_n(x) = \frac{b}{nv(x)} \sum_{k=1}^n \frac{v(X_n(s_k))|X'_n(s_k)|}{|X_n(s) - X_n(s_k)|}. \quad (7.21)$$

Subtract the simple singularity by defining

$$R_n(s, t) = \frac{v(X_n(t))|X'_n(t)|}{|X_n(s) - X_n(t)|} - \frac{v(X_n(s))}{|s - t|}.$$

For s in a fixed compact subinterval of $(0, \pi)$, this function of t has uniformly bounded variation, with at most a bounded jump at $t = s$. To verify this near s , factor $X_n(t) - X_n(s) = (t - s)D_n(s, t)$. Both D_n and X'_n are uniformly bounded away from zero there; the numerator remaining after subtraction vanishes at $t = s$, so division leaves a uniformly C^1 function on each side, multiplied by $\text{sgn}(t - s)$. Away from s , the denominator stays uniformly away from zero. Midpoint quadrature therefore has error $O(1/n)$, uniformly also when s is arbitrarily close to a quadrature point.

Changing variables $y = X_n(t)$ in the integral, and first deleting $(s - \epsilon, s + \epsilon)$, gives

$$\begin{aligned} \int_0^\pi R_n(s, t) dt &= -Lv(x) + v(x) \log \frac{1 - x^2}{X'_n(s)^2 s(\pi - s)} \\ &= -Lv(x) + v(x) [2 \log(1 - \psi'(\theta)/n) - \log(s(\pi - s))]. \end{aligned} \quad (7.22)$$

The cancellation follows because the deleted distances in the y variable are $|X'_n(s)|\epsilon + O(\epsilon^2)$ on both sides.

Write

$$\tau = \frac{ns}{\pi} + \frac{1}{2} = m + t, \quad m = \lfloor \tau \rfloor, \quad 0 < t < 1.$$

Since $(s - s_k)n/\pi = \tau - k$, midpoint quadrature in (7.21) gives explicitly

$$\lambda_n(x) = \frac{b}{\pi} \left[\sum_{k=1}^n \frac{1}{|\tau - k|} + \frac{1}{v(x)} \int_0^\pi R_n(s, t') dt' \right] + O(1/n).$$

On the compact interval under consideration, $1 \leq m \leq n - 1$ for large n , and $b = \sin(\pi t)$. Splitting the elementary harmonic sum at m gives

$$\sum_{k=1}^n \frac{1}{|\tau - k|} \leq \log(m(n - m)) + 2 + \frac{1}{t} + \frac{1}{1 - t}.$$

Also, uniformly in this compact interval,

$$\log(m(n - m)) - \log(s(\pi - s)) = 2 \log n - 2 \log \pi + O(1/n).$$

Substituting these estimates and (7.22) into (7.21), and using $b/t \leq \pi$, $b/(1-t) \leq \pi$, yields

$$\lambda_n(x) \leq b \left[\frac{2}{\pi} \log n - \frac{Lv(x)}{\pi v(x)} + \frac{2 - 2 \log \pi}{\pi} \right] + 2 + O(1/n).$$

For large n the bracket is nonnegative, uniformly on K . We may replace $b \leq 1$ by one. Since $2 + (2 - 2 \log \pi)/\pi < 3$, this proves (7.19) away from the nodes. At the nodes, $\lambda_n = 1$, and (7.19) follows for large n as well. $\square$

Remark (A numerical check of the finite term). At phase midpoints, $b = 1$, and the harmonic sum above satisfies

$$\sum_{k=1}^n \frac{1}{|m + \frac{1}{2} - k|} = \log(m(n-m)) + 2\gamma + 4 \log 2 + o(1),$$

uniformly on compact interior intervals. Thus the corrected quantity

$$E_n(x) = \lambda_n(x) - \frac{2}{\pi} \log n + \frac{Lv(x)}{\pi v(x)}$$

at these points tends to $\kappa_V = (2/\pi)(\gamma + \log(4/\pi)) = 0.521251626\dots$, the constant in Vértesi's asymptotic formula [11]. The reproducible script `checks/amplitude_numerics.py` evaluates three fixed examples $h = 1$, $1 - z/2$, $(1 - z/2)^2$ for $n = 50, 100, \dots, 800$, using the derivative-weight identity and closed expressions for Lv/v . On the prescribed phase midpoints in $K = [-0.8, 0.8]$, the largest computed values at $n = 800$ are, respectively, 0.5212524, 0.5215144, and 0.5217777.

Every positive smooth function v on I can be approximated in $C^1(I)$ by amplitudes occurring in this lemma. Approximate $1/v$ by a strictly positive real polynomial q of degree d , and form

$$J(z) = z^d q((z + z^{-1})/2).$$

This is a real polynomial, and $|J(e^{i\theta})| = q(\cos \theta) > 0$. In its factorization, replace every factor $z - r$ with $|r| < 1$ by $1 - \bar{r}z$, retaining the leading scalar. The resulting polynomial h has no zero on the closed disk and the same modulus as J on the unit circle. Conjugate roots are reflected together, so its coefficients are real; changing its sign makes $h(0) > 0$. Thus

$$|h(e^{i\theta})| = q(\cos \theta), \quad |h(e^{i\theta})|^{-1} = q(\cos \theta)^{-1}.$$

Polynomial approximation in C^1 follows by uniformly approximating the derivative and then integrating. Inversion is continuous in C^1 on functions with a positive lower bound. Hence q^{-1} approximates v in $C^1(I)$, and (7.1) gives arbitrarily accurate uniform approximation of Lv/v .

Remark (Dependence on interval length). The loss on a small interval is already visible in a simpler construction. Let $I_\ell = [-\ell/2, \ell/2]$, where $0 < \ell < 1/4$, and choose a smooth v equal to 1 on I_ℓ and to $\delta = 1/\log(1/\ell)$ outside $[-\ell, \ell]$, with $\delta \leq v \leq 1$ and $\|v'\|_\infty = O(1/\ell)$. The integral outside $[-\ell, \ell]$ and the transition regions give, uniformly on I_ℓ ,

$$Lv/v = 2 \log(1/\ell) + O(1).$$

Approximate by an amplitude as above, with error at most one in this quotient. Lemma 10 gives the upper bound in

$$\sup_{I_\ell} \lambda_n = \frac{2}{\pi} \log(n\ell) + O(1).$$

The matching lower bound follows from (7.21)–(7.22) at a phase midpoint, where $b = 1$; such points lie in I_ℓ for all large n . The constants here can be absolute, with the starting index depending on ℓ . Thus Tao's interval constant can be at least $(2/\pi) \log(1/|I_\ell|) - O(1)$ when the array changes with ℓ . The gap losses in Theorem 2 are consistent with this dependence; its additional conclusion concerns one fixed array for which every finite common constant fails.

7.5 Assembly of the array

Proof of Theorem 2. Fix $M > 0$, and carry out Subsections 7.1–7.3 with $A = \pi(M + 4)$. For each $j \geq 2$, choose an amplitude $\tilde{v}_j = |h_j|^{-1}$ as in Subsection 7.4 so close to v_j that

$$\left\| \frac{L\tilde{v}_j}{\tilde{v}_j} - \frac{Lv_j}{v_j} \right\|_\infty < 1/2. \quad (7.23)$$

Let $K_j = [-1 + 1/j, 1 - 1/j]$. Choose a strictly increasing sequence of integers N_j , each larger than $\deg h_j$ and the threshold in Lemma 10 for h_j, K_j . For $N_j \leq n < N_{j+1}$, use as row n the n roots of (7.18) with $h = h_j$. Fill the finitely many earlier rows with Chebyshev roots.

Fix $x \in (-1, 1)$. The block index j tends to infinity with n , and $x \in K_j$ for all large j . Equations (7.17) and (7.23) give $L\tilde{v}_j(x)/\tilde{v}_j(x) \geq A + 1/2$ eventually. Hence (7.19) gives, for all sufficiently large n ,

$$\lambda_n(x) \leq \frac{2}{\pi} \log n + 3 - \frac{A + 1/2}{\pi} < \frac{2}{\pi} \log n - M.$$

This proves (1.3) and the failure of an absolute additive constant.

For the final assertion, (7.16), (7.19), and (7.23) give at every fixed $x \in U$

$$\limsup_{n \rightarrow \infty} \left(\lambda_n(x) - \frac{2}{\pi} \log n \right) \leq 3 + \frac{1}{2\pi} - \frac{Lu(x)}{\pi u(x)}. \quad (7.24)$$

Choose a gap (b, c) with both endpoints in $(-1, 1)$. By (7.7), $Lu(x)/u(x) \rightarrow +\infty$ as $x \downarrow b$. Given any finite C , there is a nonempty interval $(b, b + \sigma)$ on which the right side of (7.24) is strictly less than $-C - 1$. This interval is disjoint from G_C , proving the second assertion. $\square$

8 Consequences

Corollary 11 (Positive-measure sets). *Let $E \subset (-1, 1)$ be measurable with $|E| > 0$, and let $0 < c < 2/\pi$. For all sufficiently large n , there is $x \in E$ such that $\lambda_n(x) > c \log n$.*

Proof. Otherwise $\lambda_n(y) \leq c \log n$ for every $y \in E$ and every n in an infinite set S . The argument in Section 6 applies to these rows, with all limits taken as $n \rightarrow \infty$ through S , and gives the same contradiction $|E|/A \geq |E|$ with $A > 1$. $\square$

Corollary 12 (No uniform interval constant). *For the array satisfying both assertions of Theorem 2, with any $M > 0$, there is no finite C such that*

$$\sup_{x \in I} \lambda_n(x) \geq \frac{2}{\pi} \log n - C$$

for every nondegenerate compact interval $I \subset (-1, 1)$ and all sufficiently large n , with the threshold allowed to depend on I .

Proof. Suppose that such a C exists. If G_{C+1} misses a nondegenerate compact interval $K_0 \subset (-1, 1)$, the closed sets

$$F_N = \{x \in K_0 : \lambda_n(x) \leq (2/\pi) \log n - C - 1 \text{ for all } n \geq N\}$$

cover K_0 . Baire's theorem gives an F_N containing a nondegenerate compact subinterval I . The resulting upper bound on I contradicts the assumed lower bound for large n . Thus G_{C+1} is dense, contrary to Theorem 2. $\square$

Use of generative AI

GPT-6 Astra was used to generate proofs and draft the manuscript; GPT-5.6 Sol and Claude Opus 5 were used for editorial review. The Lean formalization was generated using OpenAI Codex (GPT-6).

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